

EGERTON



UNIVERSITY

UNIVERSITY EXAMINATIONS

REGULAR -NJORO CAMPUS

SECOND SEMESTER, 2019/2020 ACADEMIC YEAR

THIRD YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

(GEOGRAPHY),), BACHELOR OF SCIENCE AND BACHELOR OF ART

GEOG 324: GEOGRAPHIC INFORMATION SYSTEM

STREAM: BSc (Geog), BSc. & BA.

TIME: 2 HRS

EXAMINATION SESSION: APRIL

YEAR: 2021

INSTRUCTIONS:

- (i) Answer question ONE and any other TWO questions
- (ii) Use sketch maps and diagrams whenever they serve to illustrate an answer
- (iii) Do not write on the question paper
- (iv) Show your working clearly

Question One

- a) What explanatory notes on the following:
 - i). Metadata **(2 Marks)**
 - ii). Attribute table **(2 Marks)**
 - iii). Raster data format **(2 Marks)**
 - iv). Vector data format **(2 Marks)**
 - v). Satellite Navigation System **(2 Marks)**
- b) Differentiate between spatial data and empirical statistical datasets. **(4 Marks)**
- c) Describe three applications of digital elevation models. **(3 Marks)**
- d) What do you understand by interpolation in GIS **(3 Marks)**
- e) Explain ways of minimizing possible errors in a GIS **(4 Marks)**
- f) Explain ways of converting analogue data format to digital data format. **(3 Marks)**

g) Explain limitations associated with the use of satellite imagery for analysis.

(3 Marks)

Question Two

a) Define ‘spatial query’ as used in GIS database systems

(2 Marks)

b) Examine the types of operators used in conditional statements applied when querying

(12 Marks)

Question Three

a) Explain the major image classification techniques.

(8 Marks)

b) Examine the application of Remote Sensing by Government of Kenya

(12 Marks)

Question Four

a) How are the following spatial analysis tools applied?

i). Reclassification

(3 Marks)

ii). Coverage rebuilding

(3 Marks)

iii). Overlay analysis

(3 Marks)

iv). Clip Operator

(3 Marks)

b) Examine the modern technological trends in Remote Sensing and GIS

(8 Marks)

Question Five

a) By use of illustrations, explain how satellite navigation systems work.

(10 Marks)

b) Examine the rationale for the development of Kenya National Spatial Data Infrastructure.

(10 Marks)
